

Python Operators – One Program

Program

```
a = 10
b = 3

# Arithmetic Operators
print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Modulus:", a % b)
print("Exponent:", a ** b)
print("Floor Division:", a // b)

# Comparison Operators
print("Equal:", a == b)
print("Not Equal:", a != b)
print("Greater Than:", a > b)
print("Less Than:", a < b)
print("Greater Than or Equal:", a >= b)
print("Less Than or Equal:", a <= b)

# Assignment Operators
x = 10
x += 5
print("x += 5:", x)

x -= 3
print("x -= 3:", x)

x *= 2
print("x *= 2:", x)

x /= 4
print("x /= 4:", x)

# Logical Operators
p = True
q = False

print("Logical AND:", p and q)
print("Logical OR:", p or q)
print("Logical NOT:", not p)

# Bitwise Operators
m = 5
n = 3

print("Bitwise AND:", m & n)
print("Bitwise OR:", m | n)
print("Bitwise XOR:", m ^ n)
print("Bitwise NOT:", ~m)
print("Left Shift:", m << 1)
print("Right Shift:", m >> 1)

# Membership Operators
numbers = [1, 2, 3, 4, 5]

print("2 in numbers:", 2 in numbers)
print("6 not in numbers:", 6 not in numbers)

# Identity Operators
c = numbers
d = [1, 2, 3, 4, 5]

print("c is numbers:", c is numbers)
print("c is not d:", c is not d)
```

Output

```
Addition: 13
Subtraction: 7
Multiplication: 30
Division: 3.3333333333333335
Modulus: 1
Exponent: 1000
Floor Division: 3

Equal: False
Not Equal: True
```

Greater Than: True
Less Than: False
Greater Than or Equal: True
Less Than or Equal: False

x += 5: 15
x -= 3: 12
x *= 2: 24
x /= 4: 6.0

Logical AND: False
Logical OR: True
Logical NOT: False

Bitwise AND: 1
Bitwise OR: 7
Bitwise XOR: 6
Bitwise NOT: -6
Left Shift: 10
Right Shift: 2

2 in numbers: True
6 not in numbers: True

c is numbers: True
c is not d: True